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Identifying low vision rehabilitation priorities using a conjoint analysis approach

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Abstract. To identify the relative importance of a range of activities amongst patients with age-related macular degeneration (AMD) and to explore the visual function and clinical characteristics associated with the different priorities for low vision rehabilitation. A prospective sample of 122 patients with AMD attending the low vision clinics at Edinburgh Eye Hospital were given several tests including: binocular visual acuity (logMAR), binocular contrast sensitivity (Pelli–Robson), quality of life (QoL) NEI-VF 25 questionnaire and utilities were determined using a time trade-off task (TTO) and a conjoint analysis task. Clinical grading of AMD severity was according to the international classification system. The relative importance of five different activities, identified by factor analysis in a previous study, was assessed in a paired comparison paradigm using discrete-choice conjoint analysis. The activities were: reading, recognising faces, getting about outside, problems associated with glare, and difficulties associated with performing household tasks. Each activity was represented at three levels of difficulty (none, a few, and a lot). Utilities derived from conjoint analysis were compared with TTO utilities. A factor analysis showed that AMD severity and basic measures of visual function were only moderately related to TTO utilities but independent of conjoint utilities. Analyses of individual patient utilities showed two distinct patient clusters; those for whom reading was of greatest importance and those for whom getting about outside was of greatest importance. Conjoint analysis offers the potential of identifying individual patient priorities for low vision rehabilitation. © 2005 Published by Elsevier B.V.

Keywords: Low vision; Rehabilitation priority; Time trade-off utility; Conjoint analysis utility; Quality of life; Age-related macular degeneration

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1. Introduction

Establishing priorities for rehabilitation is essential for two reasons, firstly to help determine what is important to the individual patient in terms of their rehabilitation needs and secondly, to provide an estimate of the value of the service as a whole to inform health care managers' decisions about its utility and cost effectiveness of the service in the context of other health care services competing for limited resources. There are many approaches to measuring importance or utility and most can be grouped into one of two categories; these are (a) direct magnitude estimation or rating scales and (b) choice-based methods, the latter of which include procedures such as standard gamble, time trade-off (TTO) and conjoint analysis. In the UK, the National Institute for Clinical Excellence (NICE) has been established to appraise the cost-effectiveness of health care interventions, including medicines, medical devices, and medical procedures [1] and eventually, no doubt, this will include an evaluation of the impact of low vision services. The recommendations of NICE are that the value or utility of health effects be assessed using choice-based methods and they specifically exclude the use of rating scales. Presumably, this recommendation of NICE assumes that the different choice-based methods will provide equivalence of utilities.

2. Purpose

One of the purposes of the present study was to compare and assess the equivalence of TTO and conjoint analysis for deriving utilities associated with vision-related quality of life (QoL) on the same cohort of patients with age-related macular degeneration (AMD). Comparisons of these utilities were also made with clinical grade of AMD, basic measures of visual function and standardised vision-related QoL questionnaire data.

3. Methods

A more detailed description of the study is given elsewhere [2]. The study comprised a prospective sample of 122 patients attending the low vision clinics at Edinburgh Eye Hospital. AMD was graded using the international classification system [3]. The four monocular grades were: mild dry, moderate dry, late dry (geographic atrophy affecting the inner macular) and wet (including disciform scar). Three binocular categories of AMD severity were derived from the monocular grades: "mild"=mild in both eyes OR mild in one eye and moderate in the other eye, "moderate"=moderate in both eyes OR mild in one eye and severe in the other eye, "severe"=late dry or wet AMD in both eyes. The visual function tests included: monocular and binocular best corrected distance and near visual acuity (logMAR) and contrast sensitivity (Pelli–Robson). QoL questionnaires included the NEI VF-25, a life-style questionnaire and the two choice-based methods for determining utility were TTO and conjoint analysis.

4. Choice-based methods for measuring importance

4.1. Time trade-off

Patients were asked to estimate the proportion of their expected remaining life years they would trade for a theoretical new technology or treatment that would restore perfect

Table 1
An example of a paired-comparison of profiles of five daily-living difficulties for two hypothetical people as used in the conjoint analysis task

Person 1	Person 2
A lot of difficulties reading and seeing fine detail	A few difficulties reading and seeing fine detail
No difficulties recognising faces across a room	No difficulty recognising faces across a room
No difficulties getting about outside alone	A lot of difficulties getting about outside alone
A few difficulties with glare from bright lights	No difficulty with glare from bright lights
A lot of difficulties with pouring liquids and household chores	A few difficulties with pouring liquids and household chores

Patients were asked to state, "Which person, in your opinion, is worse off?"

vision. To make the task easier, patients were asked which of three values best represented their remaining life expectancy, 30 years, 20 years or 10 years. They were then asked how many of these years they were prepared to trade to obtain perfect vision. Utility was calculated as a function of the percentage (P) of years traded and defined as:

$$\text{Utility} = \{(100 - P)/100\}$$

here a utility of 1 is equivalent to perfect health (i.e. will not trade any remaining life) and a utility of 0 is equivalent to death (i.e. would trade all expected remaining life).

4.2. Conjoint analysis

This method of assessing the relative importance of different attributes (e.g. daily activities) has a widely established use for evaluating health state preferences [4]. Patients were asked to consider themselves in one of two different states as a consequence of their AMD. These two states were described by paired profiles of five independent attributes of daily living identified from a previous study [5]. The profiles comprised different combinations of the five attributes described in terms of one of three levels of perceived difficulty ("no difficulty", "a few difficulties", and "a lot of difficulties"). An example paired-comparison is shown in Table 1. A fractional factorial replicate of this 3^5 design involved each patient responding to 15 paired-comparisons in which they were required to state which person, in their opinion, was worse off.

Two questionnaires were used depending on whether a patient could, or could not, read N36 comfortably. For those who could read 36-point print a 5-attribute questionnaire was used, whereas for those who were unable to read 36-point print, the questionnaire was reduced to only 3 attributes (i.e. recognising faces, getting about outside, pouring liquids and household chores) and relied on temporal rather than spatial comparisons.

5. Patients

The mean age of the patients was 77.6 years (S.D. = 6.8) and, using the binocular grade of AMD severity, 16.4% were classified as "mild", 33.3% as "moderate" and 50.3% as "severe". The mean binocular distance logMAR visual acuity was 0.50 (S.D. = 0.43), mean binocular near logMAR visual acuity 0.72 (S.D. = 0.47) and mean binocular Pelli–Robson contrast sensitivity was 1.15 (S.D. = 0.43).

6. Results

6.1. Time trade-off (TTO) utilities

The mean age of respondents' choice for one of the three life-expectancy groups of 30, 20 or 10 years was 68.5 years, 76.8 years and 81.5 years, respectively. While this is an indication of the validity of their choice, the frequency distribution of the percentage years a person would trade showed a definite quantisation for proportions of 0%, 25%, 50% and 75%. About half were not prepared to trade any of their estimated remaining life while 1 in 4 were prepared to trade 50% of their expected remaining years.

An ANOVA showed no significant difference across the three life expectance groups in terms of binocular distance and near visual acuity, although the group expecting to live the longest had significantly better contrast sensitivity than the other two groups ($F=3.7$, $df=2$, $p=0.2$). Using all respondents, binocular classification of AMD severity significantly affected TTO utilities ($F=19.8$, $df=2$, $p<0.001$) with a mean utility of 0.96 for mild AMD, 0.92 for moderate AMD and 0.69 for severe AMD. In other words, people with severe AMD were prepared to trade more of their remaining expected life for perfect vision than were those patients with mild or moderate AMD. Post hoc analyses showed no significant difference between mean utilities for mild and moderate AMD but a significant difference between moderate and severe AMD ($p<0.001$).

The proportion of patients who refused to trade any of their estimated remaining life (non-traders) was highest amongst those with mild binocular AMD severity (88%) and least amongst those with severe binocular AMD severity (30%). Correlations between TTO utilities, binocular visual function (distance VA, near VA, contrast sensitivity), AMD severity in the better eye, worse eye and binocular AMD severity classification were low, ranging from $r=0.27$ to $r=0.40$ (all $p<0.01$) when all patients were included in the analyses. When the non-traders were removed from the analyses, none of these correlations remained significant at $p=0.01$, suggesting that TTO utilities, amongst those prepared to trade, were unrelated to visual function and clinical status.

6.2. Conjoint analysis utilities

In these analyses, utilities for each attribute are determined by the *differences* in importance between the assigned values of "no difficulties", "a few difficulties" and "a lot of difficulties". From the 5-attribute conjoint task, outdoor mobility and reading were the most important. Using logit analysis, utilities were computed for each attribute in terms of the preference *difference* between the levels of "no difficulties" and "a lot of difficulties." Setting the utility for the most important attribute of "getting about outside alone" as unity, the relative utilities of the other attributes were: reading $=0.72$, pouring liquids $=0.66$, recognising faces $=0.52$ and problems with glare $=0.28$. Thus, for the sample, reading was 72% as important as getting about outside alone, while recognising faces was only about half as important. There was a similar relative importance for the common variables in the 3-attribute task completed by patients with poorer visual acuity. Using Bonferroni correction for $p<0.01$, none of the utilities generated from conjoint analysis were significantly related to measures of visual function (i.e. binocular distance or near visual acuity, or binocular contrast sensitivity) or the degree of AMD severity, and neither was there any relationship between age or gender and conjoint utilities.

Binary logistic regression analyses identified no predictors of stated priorities from clinical status, visual function or QoL difficulties. However, a factor analysis of the visual function data, AMD grading, TTO utilities, conjoint utilities and a self-rating of general health demonstrated four principal components which accounted for 71% of the variance. The first component was characterised by aspects of visual function and AMD grade, which were only moderately related to TTO utilities. The next three components characterised separate independent dimensions of the

conjoint utilities; one related to priorities for reading and seeing detail, another related to priorities for mobility and a third related to priorities for recognising faces. Further factor analyses of the associations between conjoint utilities and the NEI VF-25 QoL item scores, and also between conjoint utilities and item scores for life-style difficulties, showed a similar factor structure with the same three components of conjoint utilities that were independent of the QoL instruments and TTO utilities.

It is possible, by means of a Hierarchical Bayesian analysis, to derive individual patient utility profiles across the five stated conjoint attributes. Factor analysis of these individual utilities shows that there were two principal groups of patients' priorities and similar findings have recently been found in a sample of patients with vision loss from glaucoma. About one-third of the AMD sample had reading as their principal priority and about one-third had outdoor mobility as their principal priority.

7. Conclusion

There are limitations to the use of TTO, with many patients (i.e. mostly those with mild AMD and better visual acuity) refusing to trade. Utilities derived from TTO are only moderately related to aspects of visual function and the severity of AMD, and are also independent of conjoint analysis utilities. TTO and conjoint analysis tasks are clearly measuring different properties of importance. Conjoint analysis, by its contextual framing of options, more closely approximates an individual's practical decision making.

Factor analysis of conjoint utilities reveals two principal clusters of patients in terms of utilities and these require radically different rehabilitation approaches. Patients whose priority is for getting about outdoors alone will initially require mobility training, and patients whose priority is for reading are more likely to benefit initially from optical low vision devices. The findings emphasise the need for an integrated approach to low vision rehabilitation; to address the wrong priority for an individual is not only wasteful of resources but is likely to have a negative impact on the patient's psychological state.

The findings also demonstrate that priorities for rehabilitation cannot be predicted from clinical state, visual function or rated QoL difficulties. While rehabilitation preferences are not synonymous with need, preferences need no inferences. Conjoint analysis offers the opportunity of deriving individual preferences, although there is a need for more work to identify the best method of assessing importance in rehabilitation. Only by this means will it be possible to determine the appropriate priorities to address for an individual patient and derive the cost-effectiveness of low vision services as a whole.

References

[1] National Institute for Clinical Excellence, Guide to the Methods of Technology Appraisal, Department of Health, London, 2004 (April) (Section 5.5: Valuing health effects).
[2] P.A. Aspinall, et al. Quality of life and relative importance: a comparison of time trade-off and conjoint analysis methods in patients with age related macular degeneration (AMD). Br. J. Ophthalmol. 89 (in press).
[3] A.C. Bird, et al., International classification and grading system for age related maculopathy and age related macular degeneration, Surv. Ophthalmol. 39 (5) (1995) 367–374.
[4] S.L. Szeinbach, et al., Using conjoint analysis to evaluate health state preferences, Drug Inf. J. 33 (1999) 849–858.
[5] P. Nelson, P. Aspinall, C. O'Brien, Patients' perceptions of visual impairment in glaucoma, Br. J. Ophthalmol. 83 (1999) 546–552.